

LASER CUTTING & ENGRAVING

The complete workshop poster — beam physics · machine anatomy · safety · materials · LightBurn settings · calibration · troubleshooting

455 nm / 10.6 μm

BLUE DIODE · CO₂

0.1–0.3 mm

KERF & SPOT SIZE

300 × 300 mm

TYPICAL BED (A6 PRO)

up to 15 000 mm/min

ENGRAVE SPEEDS

Class 1 / 4

ENCLOSED / OPEN

01 🔧 HOW A LASER CUTTER WORKS

LASER = Light Amplification by Stimulated Emission of Radiation — a single-colour (monochromatic), coherent, focused beam of light.

- ▶ **Generate:** a diode (455 nm blue) or gas tube (CO₂, 10.6 μm) emits an intense beam of one wavelength.
- ▶ **Focus:** the beam is collimated then squeezed by a lens down to a tiny spot → power density explodes (W/mm²), heating material to **vaporisation / melting point**.
- ▶ **Cut** = moving the focused spot along a path, vaporising material (CNC gantry + steppers, GRBL firmware).
- ▶ **Engrave** = same beam at lower power, scanning the surface to remove only a thin layer (raster).
- ▶ **Kerf:** the width of material removed by the beam (0.1–0.3 mm) — designs must compensate for it.
- ▶ **Air assist** blows vapor & smoke away → cleaner edges, less char, fewer flames.
- ▶ **Speed ↔ power trade-off:** too fast = no cut-through; too slow = charred edges. Find the balance with test grids.
- ▶ **Focus position matters:** best cut happens at the focal point — use the focus tool / z-height every job.

Laser source

DIODE / CO₂

MIRROR

LENS

Workpiece (material) — vaporised along the cut path · kerf = removed width

Focal point — spot

Air assist blows vapor away ↓

02 🧑‍🔬 LASER TYPES — WHAT & WHY

TYPE	λ / POWER	CUTS & MARKS
CO ₂ gas	10.6 μm · 40–150 W	Wood, acrylic <i>incl. clear</i> , glass, leather, paper, fabric, stone marking. The workshop classic — clean edges on acrylic.
Blue diode 455 nm	1.5–20 W	Wood (thin), dark cast acrylic, leather, paper, anodised Al marking. Cheap, compact, air-cooled. Cannot cut clear/transparent or bare metal.
IR diode 1064 nm	~1 W head	Metal marking (steel, Ti) on coated / sprayed surfaces. Optional head for diode machines (e.g. DAJA A6 Pro).
Fiber / DPSS 1064 nm	20–100 W+	Industrial metal cutting & deep engraving (brass, steel, Al).

- ▶ **Why wavelength matters:** CO₂'s 10.6 μm is absorbed by organics & acrylics (even clear). 455 nm blue passes straight through clear materials but is absorbed by dark pigments. 1064 nm IR is absorbed by metals.
- ▶ **Household / school / hobby machines** are overwhelmingly **diode** or **small CO₂**. "20 W" marketing = 5.5–20 W real output depending on brand — always trust test cuts, not the label.

03 ⚙️ MACHINE ANATOMY (DIODE / GRBL)

ENCLOSURE — Class 1 when lid closed

X/Y gantry rail

LASER MODULE

beam ↓

Workpiece — taped flat, focus set with focus tool

Honeycomb / slat bed — lets beam pass, holds parts

FUMES

Exhaust fan + duct

CONTROLLER

GRBL · G-code · USB

→ outside

- ▶ **Steppers + belts** move the gantry; GRBL reads G-code that LightBurn generates from your vectors.
- ▶ **Safety hardware:** lid-open switch (kills beam), E-stop button, tilt sensor — never bypass them.
- ▶ **Lens & mirrors need cleaning** — smoke residue slowly dims the beam (isopropyl + lens paper).

04 🛡️ SAFETY — READ THIS FIRST

- ▶ **EYES** Never stare into the beam. 455 nm blue is **visible and can blind instantly**; CO₂ 10.6 μm is **invisible** — silent damage. Enclosed machine = Class 1 (safe with lid shut). Open machines = Class 4 — wear laser goggles rated for the wavelength (OD ≥ 4).
- ▶ **FIRE** Laser + wood = **real fire risk**. Never run unattended. Keep a fire extinguisher nearby; watch for sustained flame or heavy smoke on thick cuts and hit E-stop.
- ▶ **FUMES** Always run the exhaust. Burning plastics (PVC, ABS) release **toxic gases** — only laser-safe materials, see panel 10.
- ▶ **HOT** Freshly cut parts are **hot** — wait for them to cool, use gloves. Engraved edges can be sharp.
- ▶ **MACHINE** Keep the lid closed while running; keep the work area clear of flammable clutter; never defeat the lid switch.
- ▶ **POWER** Don't run the diode at > 90 % power for long jobs — it extends lens & diode life.
- ▶ **SETUP** Focus with the focus tool before every job; ensure the material is flat and taped down.

Golden rule: if you don't know exactly what the material is — **don't laser it**. Test a corner scrap first.

05 🗺️ MATERIALS MATRIX — WHAT LASERS CAN DO

MATERIAL	CUT	ENGRAVE	BLUE DIODE 455 NM NOTES
Plywood / basswood	✓	✓	Best all-rounder. 3 mm easy; ≥6 mm needs 2–3 passes. Watch veneer char.
Hardwood (oak, walnut)	△	✓✓	Engraves beautifully; cuts slowly on diode (resins). Great on CO ₂ .
MDF	✓	✓	Clean cut, dusty; edges darken. Vent well.
Balsa	✓✓	✓	Easiest wood — very fast cuts, low power.
Acrylic — dark cast	✓	✓	Cut cast only; flame-polished edges. Keep protective film on.
Acrylic — clear / extruded	✗	✗	455 nm passes straight through; extruded melts ugly. CO ₂ handles both.
Leather (real / PU)	✓	✓	Fumes smell — vent. PVC pleather = banned (chlorine gas).
Glass	✗	with prep	Needs black paint / colour sheet / wet-towel backing, then engrave.
Anodised aluminium	✗	✓✓	Instant crisp white mark (~15 000 mm/min @ 90 %). Bare metal bounces the beam.
Stainless / brass / copper	✗	spray	Only with marking spray/paste or 1064 nm IR head; multiple passes.
Stone / ceramic tile	✗	✓ CO ₂	Diode marks poorly; CO ₂ engraves well.
Paper / cardboard	✓	✓	Low power, fast — fire risk if too slow. Great for prototypes.
Fabric (cotton, wool)	✓	✓	Seals edges. Synthetics melt — avoid.
Cork	✓	✓	Easy, smells toasted.

Key: ✓ yes · △ difficult/conditional · ✗ no. CO₂ machines are more capable on transparent materials & stone; blue diodes are pigment-hungry.

07 🔄 LIGHTBURN WORKFLOW

- ▶ **31 output layers** (0–30) — colour = layer = cut parameters. **T1/T2 are TOOL layers: never fire, no parameters** — use for guides, material frame, notes.
- ▶ **Line mode** = cut vectors (speed, power, passes). **Scan mode** = engrave fills (line interval / density). **Image mode** = photos (dithering: Jarvis or Atkinson for cleanest results).
- ▶ **Max Power = Power** gives constant-power output (right for cuts). Lowering Max Power enables dynamic shading for grayscale engraving.
- ▶ **Tab**s hold parts in place: auto width = 1.5–2 × material thickness (5–8 mm for 3 mm ply), spacing 30–50 mm. Tab Cut Power **0 % = bridge stays solid**; >0 % = pulsed score.
- ▶ **File prep:** import SVG/DXF, check scale (mm!), strokes → hairline (0.001 mm) so vectors cut rather than fill, assign colours to layers.
- ▶ **Preview first:** the simulation shows order + estimated time. Then **Frame** the job to confirm it fits on the material.
- ▶ **Offset tool** applies kerf compensation (offset outside cut path by kerf/2) for press-fit parts.

Routine: Import → scale check → layer colours → Material Library settings → Preview → Frame → exhaust on → run supervised.

08 📐 DESIGN & JOINERY FOR LASER

- ▶ **Kerf compensation:** beam removes 0.1–0.3 mm — a "3 mm" slot is really ~2.8 mm. Offset cut lines by kerf/2 for press-fit joints.
- ▶ **Box joints / finger joints:** tab width = material thickness × 1 (e.g. 3 mm fingers on 3 mm ply); subtract kerf from every slot. Tolerances: laser = ±0.1 mm, much tighter than hand tools.
- ▶ **Tab**s on **nested parts** stop cut-out pieces drifting into the beam path mid-job.
- ▶ **Engrave first, cut last** — if the part moves, the engrave is already done. One file, two layers: Scan layer then Line layer.
- ▶ **Images:** use 300 dpi+, high contrast, convert to grayscale; dither with Jarvis/Atkinson for photos.
- ▶ **Text:** outline/convert text to curves before cutting (font substitution breaks files); keep cut text ≥ 10 pt.
- ▶ **Score lines:** low-power Line pass (fold lines, bend points) — cut only partway through.
- ▶ **Nesting:** pack parts tightly (shared edges save 10–30 % material) and share a single outer cut.
- ▶ **Inkscape / Fusion 360 / Onshape** all export SVG/DXF; LightBurn imports them directly.

09 🧪 CALIBRATION TEST GRID — ALWAYS

Every new material or new roll/board = new test. Speeds on the internet vary 2–3x — **your machine decides**.

ENGRAVE — 10 × 10 grid

- ▶ 10 squares × 10 rows: power 10 → 100 %, speed ladder 0.08 interval.
- ▶ Pick the darkest clean fill at acceptable speed.

CUT — 6-line ladder

- ▶ 6 lines at 100 % power: e.g. 1200 · 900 · 700 · 500 · 350 · 250 mm/min.
- ▶ Push the pieces apart — first line that drops through cleanly = winner.

Judge by: full clean cut · minimal char · crisp edge detail. Then **LightBurn** → **Material Library** → **New from Layer** to save the preset with a name like "3mm ply — cut".

Re-test after: changing focus distance, cleaning the lens, or buying a new sheet (thickness/glue vary by batch).

10 🚫 BANNED MATERIALS — NEVER LASER

MATERIAL	WHY IT'S BANNED
PVC / vinyl / pleather	Chlorine gas — toxic to you, corrodes the machine's optics . Ruins the laser.
ABS / polycarbonate (Lexan)	Melts, catches fire, releases toxic fumes.
Fiberglass / carbon fibre	Epox/resin fumes are toxic; fibres irritate.
HDPE / milk jugs	Melts, drips, floats up into the lens and fires.
Polystyrene foam	Melts and burns almost instantly.
Bare copper / brass / mirrors	Reflect 455 nm — beam bounce can hit eyes or the machine.
Unknown materials	If you don't know what it is — test a corner scrap first or skip.
Wet materials	Steam can pop/shatter surfaces (esp. wood & glass).

Check adhesive/backing too — some "laser-safe" sheets have PVC-based glue or HTV backing. Read the SDS.

11 ✅ STEP-BY-STEP WORKFLOW

1. **Design** in LightBurn, Inkscape or Fusion 360 — SVG/DXF, mm units, kerf applied.
2. **Prep the bed** — clean honeycomb, place material flat, tape down edges.
3. **Focus** with the focus tool (or set z-height) — the #1 cause of bad cuts.
4. **Import** the file, check scale, assign stroke colours → layers.
5. **Pick settings** from the Material Library or a fresh test grid.
6. **Preview** the simulation — confirm order, time, and that it fits.
7. **Frame** the job — the laser traces the bounding box on the material.
8. **Enable** exhaust + air assist. Check the fire extinguisher is close.
9. **Test** cut a corner scrap first on a new material/settings.
10. **Run and supervise** — watch for flames/smoke, E-stop if worried.
11. **Cool & remove** — parts are hot; lift with tweezers/gloves.
12. **Post-process** — peel film, sand edges, clean with isopropyl, assemble.

12 🐛 TROUBLESHOOTING

SYMPTOM	CAUSE	FIX
Not cutting through	Too fast · low power · out of focus	Slow down, add passes, refocus
Charred / black edges	Too slow · too much power · no air assist	Speed up, cut power, turn air on
Uneven cut depth	Material not flat · focus off	Tape down, refocus, shim
Engrave too light	Low power · too fast · wrong focus	Raise power, slow down, refocus
Acrylic melted & ugly	Extruded (not cast) · no air assist	Use cast acrylic + air assist
White / blurry engrave	Focus too high above surface	Lower z / refocus
Machine skips / jerks	Acceleration too high	Reduce speed/accel in LightBurn
Smoke in room	Exhaust blocked / off	Check duct & fan, vent outside
Beam getting weaker	Lens/mirror coated with residue	Clean with isopropyl + lens paper

Still stuck? Test grid beats guessing — one 5-minute test saves an hour of failed parts.